

Week 3 Report: Literature Review (final part) and Data Preprocessing (start)

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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What I did this week

This week I finished the reading part of the literature review and started preprocessing. I added four recent papers on banking fraud and intrusion detection to my review: Sanda et al. (2026), Dhammayogi and Pramatha (2026), Baisholan et al. (2025) and Pospichal et al. (2026). All of them find tree ensembles such as Random Forest and XGBoost at or near the top, which matches my plan.

For preprocessing I parsed the date column into year, month, day and day of week, and split the description field into a channel (for example NETTXN, IMPS, PCA) and the merchant. I also added the log of the amount and the ratio of amount to balance, because amounts vary over several orders of magnitude. The 33,611 rows had no missing values, so no rows were dropped. Finally I created one stratified 80/20 split with a fixed seed (42): 26,888 training rows and 6,723 test rows. Every later experiment uses this split.

Step	What I did
Dates	Parsed dd-mm-yyyy; added year, month, day, day of week
Description	Split into channel and merchant; one-hot encoded
Amounts	Single amount column, log amount, amount to balance ratio
Scaling	Standard scaling fitted on the training split only
Split	Stratified 80/20, seed 42

Next week

Finish the exploratory analysis of the cleaned data and look at how the suspicious rate changes by channel.